

Medium Access Control

The problem:

Multiple senders compete for a common, shared resource, i.e. want to use the same frequency, at same time, in the same space.

How can we resolve this?

Basics of wireless access control:

1) What is the responsibility of the Media Access Control sublayer in communication systems?

The Media Access Control (MAC) sublayer is responsible for controlling access to the physical transmission medium in a communication system.

Addressing, Frame delimiting, Medium Access control, Error detection

2) What are the new problems wrt MAC in wireless systems?

Channel access connection, Hidden Node problem, Exposed node problem, channel fading and interference.

3) What are the causes of the problems in wireless systems?

Limited bandwidth, Interference, Propagation effects, Mobility, Hidden Node problem, Exposed node problem

4) Why does CSMA/CD (Carrier Sense Multiple Access/ Collision Detection) known from Ethernet fail?

Limited Scalability, Half duplex operation, Inefficiency with modern technology, Increased Complexity, Standardization of ethernet

5) What is the hidden terminal problem? When does it happen? What happens?

The hidden terminal problem is a phenomenon that occurs in wireless networks where two or more nodes are unable to directly detect each other's transmissions due to being out of range or obstructed by obstacles, but they can communicate with a common access point or base station.

When nodes are spread out over a large area or where there are obstacles such as buildings or walls that obstruct direct communication between nodes.

Suppose Node A wants to communicate with the access point (AP), and Node B also wants to communicate with the same AP, but Node A and Node B are out of range of each other. Both Node A and Node B can successfully transmit to the AP because they are within range of the AP. However, since Node A and Node B cannot directly sense each other's transmissions, they may transmit simultaneously, causing a collision at the AP. The AP detects the collision and requests retransmissions from both nodes, leading to inefficiency and degraded performance in the network.

Various techniques and protocols, such as RTS/CTS (Request to Send/Clear to Send) mechanism and carrier sensing, are employed to mitigate the effects of the hidden terminal problem.

6) What is the exposed terminal problem? When does it happen? What happens?

the exposed terminal problem occurs when a node abstains from transmitting data due to a false belief that its transmission would interfere with another ongoing transmission, even though no actual interference would occur.

The exposed terminal problem occurs when a wireless node refrains from transmitting data even though it could do so without causing interference with other transmissions.

Let's consider an example. Suppose there are three nodes in a wireless network: Node A, Node B, and Node C. Node B wants to transmit data to Node C, but it refrains from doing so because it detects transmissions from Node A, which is communicating with a different node elsewhere in the network. However, Node B's transmission would not actually interfere with Node A's communication. As a result, Node B unnecessarily delays its transmission, leading to reduced network efficiency and throughput.

7) What is the near-far problem? What happens there?

In the near-far problem, a node that is physically closer to the receiver may transmit at a higher power level compared to a node that is farther away. As a result, the receiver may receive the strong signal from the nearby transmitter while simultaneously being unable to detect the weaker signal from the distant transmitter.

Consider a scenario where there are two transmitting nodes, Node A and Node B, and a single receiving node, Node R. Node A is closer to Node R than Node B. When both Node A and Node B transmit simultaneously, Node R may receive the signal from Node A with sufficient strength to successfully decode the data. However, the signal from Node B, being weaker due to the longer distance, may not be detected by Node R or may cause significant interference, leading to communication errors or packet loss.

8) What are the dimensions in which we can separate the access?

We can separate the access in the following dimensions:

- Temporal Separation
- Frequency separation
- Spatial separation
- Code separation
- Contention-based access

Multiplexing and Duplexing:

1) What is simplex communication? What are example systems for this?

Simplex communication is a unidirectional mode of communication where data flows in only one direction, from the transmitter to the receiver, with no feedback loop or ability for the receiver to respond directly to the sender.

TV Broadcast
Radio Broadcast
Sensor Networks

2) What are the pros and cons of TDD over FDD?

Time-Division Duplexing (TDD) and Frequency-Division Duplexing (FDD) are two duplexing techniques used in wireless communication systems to allow simultaneous transmission and reception of data.

TDD pros:

Flexibility, Spectral efficiency, Lower cost, Simple hardware

TDD Cons:

Interference, Requires Accurate Synchronization, Latency

Pros of FDD:

Improved Isolation, Predictable Performance, Suitable for Symmetric traffic

Cons of FDD:

Spectrum Efficiency, Increase Complexity, Higher cost

3) For which of the given cases is FDD preferable? Why?

Symmetric traffic Patterns, Quality of service requirement, Interference Prone environment, stability and predictability, regulatory requirements.

4) What does uplink and downlink mean?

Uplink:

The uplink is the direction in which data travels from a mobile device, such as a smartphone or tablet, to the base station or cell tower.

Downlink:

The downlink is the direction in which data travels from the base station or cell tower to the mobile device

The Division Multiple Access:

1) What is the idea of ALOHA?

The key idea of ALOHA is as follows:

Random Access: In ALOHA, each user (or node) has the freedom to transmit data whenever it has a packet ready to send

Collision Detection: When a node transmits a packet, there is a possibility of collision if another node also transmits at the same time

Collision Resolution: If a collision occurs, the nodes involved detect the collision through a lack of acknowledgment (ACK) from the receiver. Upon detecting a collision, the transmitting nodes wait for a random amount of time before attempting to retransmit their packets.

2) Why does slotted ALOHA outperform ALOHA with respect to throughput?

Slotted ALOHA outperforms pure ALOHA in terms of throughput due to its synchronized time slot structure, which reduces the likelihood of collisions and improves channel utilization.

Slotted ALOHA's synchronized time slot structure reduces collision probability, improves channel utilization, simplifies collision resolution, and enhances scalability, resulting in higher throughput compared to pure ALOHA.

3) What does implicit reservation mean?

Implicit reservation refers to the allocation of resources for data transmission without explicitly reserving them in advance. Instead, resources are dynamically allocated as needed based on the characteristics of the communication protocol or network architecture. This approach is commonly used in contention-based access methods like Carrier Sense Multiple

Access (CSMA), where nodes contend for access to the communication medium without prior reservation.

4) How does explicit reservation work?

Explicit reservation involves the process of reserving resources in advance for data transmission. This is typically achieved through signaling or reservation protocols where nodes explicitly request and allocate resources before initiating data transmission.

Request → Reservation → Acknowledgement → Data transmission → release

5) What is the difference between random access and reservation-based access? What are the pros and cons of the two strategies?

In random access, multiple nodes contend for access to the communication medium without prior coordination. Nodes may attempt to transmit data whenever they have packets to send, and collisions may occur if multiple nodes transmit simultaneously.

Pros:

Simple and decentralized, Flexibility

Cons:

Inefficiency, Lack of service quality

In reservation-based access, nodes explicitly reserve resources (such as time slots, frequency channels, or buffer space) before initiating data transmission. Reservations are typically made through signalling or reservation protocols, and nodes are allocated resources based on their requests.

Pros:

High Service quality, reduced collisions

Cons:

Overheads can reduce efficiency, Complexity

6) Compare the access latencies of both access schemes. Consider the latency under low and high loads.

Random access schemes typically experience lower access latencies under low loads but suffer from significantly increased access latencies under high loads due to frequent collisions and contention

Reservation-based access schemes, on the other hand, maintain relatively consistent and low access latencies regardless of the network load, making them more suitable for applications with strict latency requirements.

7) Compare the two access strategies with respect to the maximum load they can handle.

Reservation-based access schemes generally have a higher maximum load-handling capacity compared to random access schemes. By reserving resources in advance and minimizing contention, reservation-based access networks can maintain performance and throughput even under high load conditions.

Random access networks, on the other hand, experience performance degradation and reduced channel utilization as the load increases, limiting their maximum capacity

8) Name the limits of distributed approaches to media access?

Collision Probability, Inefficiency under high load, Lack of service quality, Vulnerability to hidden terminal problems, Limited scalability

9) How can the hidden-terminal problem be solved? Does the RTS/CTS scheme (MACA) solve the problem? In any case?

It can be solved by RTS/CTS scheme.

Here's how:

- In the RTS/CTS scheme, before a node transmits a data packet, it sends a Request to Send (RTS) frame to the intended recipient.
- Upon receiving the RTS frame, the recipient responds with a Clear to Send (CTS) frame, indicating that it is ready to receive the data packet
- Other nodes within the transmission range of the sender and recipient overhear the RTS and CTS frames and defer their transmissions, thus reducing the likelihood of collisions

Code division Multiple Access:

1) What is the trick with CDMA to avoid collisions?

The trick with Code Division Multiple Access (CDMA) to avoid collisions lies in the unique spreading codes assigned to each user. CDMA allows multiple users to transmit simultaneously over the same frequency band by assigning a unique spreading code to each user. These spreading codes have special properties that enable the receiver to distinguish between the signals of different users, even when they overlap in frequency and time.

2) Why is power control important for the correct functioning of CDMA?

Power control is crucial for the correct functioning of Code Division Multiple Access (CDMA) due to several reasons:

- Near far problem mitigation
- Maximizing system capacity

- Battery Conservation
- Coverage Extension

Power control is essential for the correct functioning of CDMA systems as it helps mitigate the near-far problem, maximize system capacity, conserve battery power, and extend coverage, ensuring efficient and reliable communication for all users

3) What happens if the keys are not orthogonal?

Following issues can occur if keys are not orthogonal:

- Interference
- Cross talk
- Capacity Reduction
- Performance degradation

4) What is the advantage of CDMA with respect to the handling of noise?

One of the key advantages of Code Division Multiple Access (CDMA) is its robustness in handling noise and interference. CDMA offers several benefits in this regard.

- Spectrum Spreading Technique
- Interference Rejection
- Soft capacity limit
- Improved signal to noise ration(SINR)

5) Does CDMA require the same, higher or lower SINR for correct transmission? Why?

CDMA typically requires a higher Signal-to-Interference-plus-Noise Ratio (SINR) for correct transmission.

This is because of the spread spectrum technique used in CDMA, which spreads the signal over a wide bandwidth, making it less susceptible to interference but requiring a higher SINR for correct decoding.